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Comparison of low-level polycyclic aromatic hydrocarbons in sediment revealed by Soxhlet extraction, microwave-assisted extraction, and pressurized liquid extraction

Nobuyasu Itoh*, Masahiko Numata, Yoshie Aoyagi, Takashi Yarita

National Metrology Institute of Japan (NMIJ), National Institute of Advanced Industrial Science and Technology (AIST),
1-1-1 Umezono, Tsukuba, Ibaraki 305-8563, Japan

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ABSTRACT

We analyzed polycyclic aromatic hydrocarbons (PAHs) present in lake sediment at low levels ($<1 \mu\text{g g}^{-1}$) by using Soxhlet extraction (Soxhlet), microwave-assisted extraction (MAE), and pressurized liquid extraction (PLE) in combination with gas chromatography and isotope-dilution mass spectrometry. Although all extraction techniques showed good repeatability for five target PAHs (relative standard deviation $<5.2\%$, $n=3$), the results decreased in the order $\text{PLE} > \text{MAE} > \text{Soxhlet}$. Differences in the results originated mainly from differences in the extraction efficiencies of the techniques for native PAHs, because all techniques gave comparable recovery yields of corresponding ^{13}C -labeled PAHs (^{13}C -PAHs) (51–84%). Since non-negligible amounts of both native PAHs and ^{13}C -PAHs were re-adsorbed on matrix in MAE, not only recovery yields of ^{13}C -PAHs but also efficiencies of extraction of native PAHs should be examined to evaluate the appropriateness of any analytical procedures.

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1. Introduction

Polycyclic aromatic hydrocarbons (PAHs), some of which are carcinogenic [1], are formed through incomplete combustion in anthropogenic processes, such as burning of fossil fuels, and natural processes, such as forest fires [2]. Because the emitted PAHs are transported for long distances, PAHs are found in most environmental samples throughout the world, including some recognized as being less affected by

human activities [3–5]. It is therefore necessary to monitor the levels of emitted PAHs not only in highly contaminated samples, but also in less-contaminated environmental samples.

To determine PAHs in sediment samples, the PAHs must first be extracted from the matrices with organic/inorganic solvents. Soxhlet extraction (Soxhlet) and ultrasonic extraction (USE), the conventional extraction techniques, have been applied to the extraction of PAHs for a long time. Recently

* Corresponding author. Tel.: +81 29 861 4318; fax: +81 29 861 6865.

E-mail address: nobuyasu-itoh@aist.go.jp (N. Itoh).

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developed techniques, such as supercritical fluid extraction (SFE), pressurized liquid extraction (PLE), and microwave-assisted extraction (MAE) have also been used to extract PAHs [6–11]. Usually, new extraction techniques are evaluated by comparison with conventional techniques such as Soxhlet and USE [8,12–14]. If the concentrations of PAHs obtained using the new techniques are comparable to those obtained using Soxhlet, it can be concluded that the new technique is more economical than Soxhlet if it is faster and it consumes less organic solvent. Recently, some of these new techniques have been applied in the certification of reference materials [15,16].

Some studies have been made of the comparative efficiencies of techniques used for the extraction of samples contaminated by PAHs [7,10]: these previous studies concluded that differences between the conventional and the novel techniques are small. However, these comparisons were performed only for high-level samples ($>1\mu\text{g g}^{-1}$), and the extrapolation of results obtained by using high-level sediments to the case of low-level sediments ($<1\mu\text{g g}^{-1}$) could be unreliable. In fact, we previously reported that the behavior of stable isotope-labeled PAHs is different from that of native PAHs during analytical processes by using the low-level sample ($<1\mu\text{g g}^{-1}$), although such differences are negligible in analyses of high-level samples [17]. Thus, the observed values of PAHs in extracts obtained from solid samples containing at low levels might also depend on the choice of extraction technique.

There are some reports of the evaluation of differences among the results of extraction techniques applied to sediment samples with low-level contaminants. Berset et al. reported that there were no significant differences among Soxhlet, USE, KOH-digestion, mechanical shaking, SFE, and PLE of 16 low-level PAHs ($<1\mu\text{g g}^{-1}$) in sediment [6]. However, it was difficult to identify any differences due to the techniques, because no common behavior was found among the 16 PAHs examined. On the other hand, Martinez et al. examined the efficiencies of Soxhlet and USE of sediment samples spiked with PAHs at 20 ng g^{-1} [8]. They applied USE in combination with an optimized clean-up procedure to validate the analytical procedure with certified reference material (sediment). However, their results were clearly lower than the certified values due to lower efficiency of extraction of native PAHs, especially for heavier PAHs such as benzo[k]fluoranthene (36% lower than the certified value) and dibenz[a,h]anthracene (50% lower than the certified value). Because the strengths of adsorption of PAHs to soil continue to change even after more than 500 days [18], it is difficult to extrapolate the results obtained using spiked sediments to non-spiked sediments. Therefore, differences in the values obtained by different techniques must be evaluated by using low-level, non-spiked sediment samples.

In this study, we examined whether there are significant differences between Soxhlet, PLE, and MAE for five PAHs (phenanthrene, fluoranthene, benz[a]anthracene, benzo[a]pyrene and benzo[ghi]perylene) in a low-level sediment sample. Factors affecting differences in the observed values and recovery yields will also be discussed.

2. Experimental

2.1. Samples

The sediment used in this study was collected from a lake in Japan as a candidate for a Japanese reference material for environmental analysis. The sediment was air-dried, pulverized, sieved ($<106\text{-}\mu\text{m}$ particles were used as the sample), homogenized, and sub-sampled into 60-g portions by Environmental Technology Service (Fukuoka, Japan). The bottled samples were sterilized by γ -ray irradiation with ^{60}Co and stored at 4°C until required. The amount of water in the sediment sample was determined by heating the sediment at 105°C for 6 h and calculated to be 2.3%. The organic carbon content was determined to be 1% by using an EA1110 elemental analyzer (CE Instruments, Milan, Italy) after removal of carbonates [19]. The particle-size distribution of the sample was determined by laser scattering using a Laser Micron Sizer (LMS-30, Seishin, Tokyo, Japan) (30% of the total amount was $<5\text{ }\mu\text{m}$, 50% was $<10\text{ }\mu\text{m}$, and 95% was $<70\text{ }\mu\text{m}$): therefore, most of the sediment sample consisted of clay and/or silt, rather than sand and organic compounds.

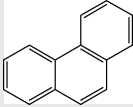
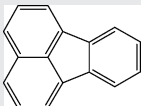
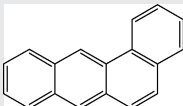
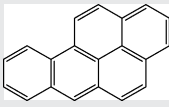
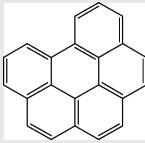
2.2. Reagents

Phenanthrene (Phe, 99.5% purity) was obtained from Supelco (Bellefonte, PA, USA). Fluoranthene (Flu, 99.8% purity), benz[a]anthracene (BaA, 99.6% purity), and benzo[a]pyrene (BaP, 99.3% purity) were obtained from Tokyo Chemical Industry (Tokyo, Japan). Benzo[ghi]perylene (BghiP, 98.3% purity) was obtained from AccuStandard (New Haven, CT, USA). Their characteristics and structures are summarized in Table 1 [2,20]. ^{13}C -labeled PAHs dissolved in *n*-nonane ($100\mu\text{g mL}^{-1}$), such as $^{13}\text{C}_6$]phenanthrene ($^{13}\text{C}_6$ -Phe, 99% isotope purity), $^{13}\text{C}_6$]fluoranthene ($^{13}\text{C}_6$ -Flu, 99% isotope purity), $^{13}\text{C}_6$]benz[a]anthracene ($^{13}\text{C}_6$ -BaA, 99% isotope purity), $^{13}\text{C}_4$]benzo[a]pyrene ($^{13}\text{C}_4$ -BaP, 99% isotope purity), and $^{13}\text{C}_{12}$]benzo[ghi]perylene ($^{13}\text{C}_{12}$ -BghiP, 99% isotope purity) were obtained from Cerilliant (Round Rock, TX, USA). Crystalline $^{12}\text{H}_{12}$]perylene (Per- d_{12} , 98% isotope purity) was obtained from the same source and used as a syringe spike. Toluene, acetone, *n*-hexane, dichloromethane (DCM), ethyl acetate (EtAc), and anhydrous Na_2SO_4 were of pesticide- and PCB-analysis grade (Kanto Chemical, Tokyo, Japan).

2.3. Preparation of calibration and surrogate solutions

Each of the native PAHs was dissolved in toluene at $\sim 200\mu\text{g g}^{-1}$, and the resulting solutions were combined. A syringe-spike solution for the calculation of the recovery yield was prepared by dissolving crystalline Per- d_{12} in toluene at $\sim 200\mu\text{g g}^{-1}$. All ^{13}C -PAH solutions were combined and part of the combined ^{13}C -PAH mixture was twenty times diluted with toluene to give a surrogate solution (for addition of $80\mu\text{L}$ per 1-g sample; 21 ng for $^{13}\text{C}_6$ -Phe, 22 ng for $^{13}\text{C}_6$ -Flu, 10 ng for $^{13}\text{C}_6$ -BaA, 6.3 ng for $^{13}\text{C}_4$ -BaP, and 7.5 ng for $^{13}\text{C}_{12}$ -BghiP). The remainder of the combined ^{13}C -PAH mixture was added to the mixture of native PAHs to form 4-level calibration solutions (correlations of peak area ratio to con-

Table 1 – Characteristics and structures of PAHs used in this study

Name	Abbreviation	Boiling point (°C) ^a	logK _{ow} ^b	Chemical structure
Phenanthrene	Phe	339	4.52	
Fluoranthene	Flu	375	5.20	
Benz[a]anthracene	BaA	435	5.91	
Benzo[a]pyrene	BaP	495	6.35	
Benzo[ghi]perylene	BghiP	525	6.90	

^a Data obtained from Finlayson-Pitts and Pitts Jr. [2].
^b Octanol–water partitioning coefficient [20].

centration ratio between 0.5 and 2: $r^2 > 0.999$ for the five PAHs).

2.4. Soxhlet extraction procedure

A 5-g portion of the sediment sample was mixed with 5 g of anhydrous Na₂SO₄ and placed in a cellulose thimble (28-mm i.d. × 100-mm long, Advantec, Tokyo, Japan). The 400 µL surrogate solution was then added. To completely remove any toluene originating from the surrogate solution, the thimble was kept overnight at room temperature in darkness before performing the extraction. The thimble was loaded on an automated Soxhlet extraction system B-811 (BÜCHI, Flawil, Switzerland) and extracted with 150 mL of *n*-hexane/acetone mixture (1:1 by volume, Hex/Ace), dichloromethane (DCM), or ethyl acetate (EtAc) for 4 h. After selecting Hex/Ace as the extraction solvent, extraction times of 8, 16, or 24 h were examined. Soxhlet extractions were performed in triplicate.

The extract was transferred to a glass vial and a small amount of activated copper powder was added to remove any elemental sulfur [22]. After filtration with a PTFE membrane filter (0.1-µm pore), the extract was concentrated by using a rotary evaporator and under a stream of nitrogen gas followed by the replacement of solvent to toluene. The concentrated extracts (dissolved in toluene) were cleaned up by using a silica-gel solid-phase extraction (SPE) cartridge (500 mg/3 mL Isolute SILICA, International Sorbent Technology, Hengoed, UK) with a Zymark (Hopkinton, MA, USA) RapidTrace automation system. PAHs were eluted with 6 mL of *n*-hexane. The

hexane was evaporated and replaced with toluene after the addition of the syringe spike (final volume: 100 µL). Recovery yields of ¹³C-PAHs (spiked at comparable levels to those in the sediment extracts) during the cleanup and concentration steps were 56% for ¹³C₆-Phe, 64% for ¹³C₆-Flu, 77% for ¹³C₆-BaA, 94% for ¹³C₄-BaP, and 102% for ¹³C₁₂-BghiP [21].

The blank samples of all the extractions performed in this study were prepared by the same procedures as used for the sediment extracts, except for the omission of the sediment sample and the surrogate solution.

2.5. Microwave-assisted extraction procedure

Microwave-assisted extraction was performed by using a MarsX system (CEM, Matthews, NC, USA). A 5-g aliquot of the sediment was weighed into a glass vessel and 400 µL of the surrogate solution was added. The mixture was kept overnight at room temperature in darkness until the extraction was performed at 1200 W under ramp-to-temperature mode (ramp time: 10 min). Three extraction solvents were tested – 30 mL of DCM, EtAc, or DCM/EtAc mixture (1:1 by volume) – at 100 °C for 30 min. DCM/EtAc was selected as the best extraction solvent (see Section 3.2), and extraction was performed at 150 °C for 10, 20, or 30 min. After cooling of the extraction vessel to room temperature, the extracts were filtered through a glass filter (GA-200, Advantec), and the resulting filtrates were treated as described above for the Soxhlet extraction procedure. MAE were also performed in triplicate.

2.6. Pressurized liquid extraction procedure

Pressurized liquid extraction was performed as reported previously [22]. In summary, PLE was performed by using a Dionex (Sunnyvale, CA, USA) ASE 200 system equipped with stainless-steel extraction cells with a volume of 11 mL. A cellulose filter was placed at the bottom of the extraction cell and a 5-g sediment sample and a few grams of anhydrous Na_2SO_4 were added. The mixture was thoroughly shaken and the 400 μL surrogate solution was added followed by additional anhydrous Na_2SO_4 to fill the vacant space in the cell. The cell was kept overnight at room temperature in darkness before performing the extraction. The sediment samples were extracted with toluene under static condition (100 or 150 °C under 15 MPa) for 10 min. After investigation of the number of extraction cycles (see Section 3.3), extraction step was performed twice. The resulting extracts were treated as described above for the Soxhlet extraction procedure. PLE were also performed in triplicate.

2.7. GC–MS analysis

GC–MS analysis was carried out with 6890/5975 equipment from Agilent Technologies (Palo Alto, CA, USA) equipped with a DB-17MS capillary column (30-m long $\times$ 0.25-mm i.d., 0.25- μm film thickness; J&W Scientific, Folsom, CA, USA). The GC system was operated in splitless mode and 1- μL portions of the extracts were injected by using an autosampler. Both the injection liner, which contained deactivated glass wool for splitless injection (Agilent Technologies), and the transfer line were maintained at 300 °C. The oven temperature was programmed to rise from 50 °C (2 min hold) to 240 °C at a rate of 10 °C min^{-1} and then to 300 °C (10 min hold) at 1.25 °C min^{-1} . Helium was used as the carrier gas at a flow rate of 1.0 mL min^{-1} in the constant-flow mode. The electron-impact (EI) ionization energy was 70 eV. Data for quantification were obtained in the selected-ion monitoring (SIM) mode. The ions selected for each PAH (m/z) were as follows: Phe (178.1, 176.1), Flu (202.1, 200.1), BaA (228.1, 226.1), BaP (252.1, 250.1), BghiP (276.2, 277.2), $^{13}\text{C}_6$ -Phe (184.1, 182.1), $^{13}\text{C}_6$ -Flu (208.1, 206.1), $^{13}\text{C}_6$ -BaA (234.2, 232.2), $^{13}\text{C}_4$ -BaP (256.2, 257.2), $^{13}\text{C}_{12}$ -BghiP (288.2, 289.2), and Per- d_{12} (264.2, 265.2). The underlined ions were used for quantification. The PAH concentrations were calculated by comparison of peak areas using the internal-standard technique.

3. Results and discussion

3.1. Investigation of solvents and times for Soxhlet extraction

Soxhlet extraction (Soxhlet) is a well-known conventional extraction technique not only for PAHs, but also for other organic pollutants, such as pesticides and polychlorinated biphenyls (PCBs) [15,16,23]. It is also used as a reference extraction technique in evaluations of the efficiency of new extraction techniques [11,12,14].

In Soxhlet extraction, both the solvent and the time for extraction have significant effects on the efficiency of extrac-

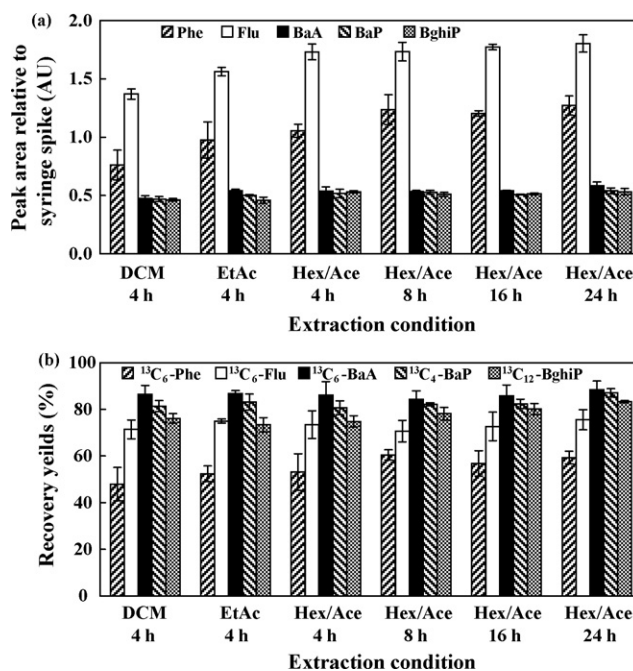


Fig. 1 – Efficiencies of extraction of native PAHs (a) and recovery yields of ^{13}C -PAHs (b) by Soxhlet extraction (DCM for 4 h, EtAc for 4 h, or Hex/Ace for 4, 8, 16, or 24 h). Error bars indicate the standard deviation ($n = 3$).

tion of target compounds [23]. To achieve better efficiencies of extraction of PAHs from the sediment sample, we first optimized the solvent for extraction. Fig. 1 shows the efficiencies of extraction of native PAHs (a) and the recovery yields of ^{13}C -PAHs (b) with Hex/Ace, DCM, and EtAc for 4 h and with Hex/Ace for 8, 16, and 24 h. Hex/Ace showed comparable extraction efficiencies to those of the other solvents, or better, for all native PAHs (from +9% to +28% relative to DCM and from –1% to +14% relative to EtAc). On the other hand, recovery yields of ^{13}C -PAHs with Hex/Ace were closer to those achieved with other solvents (from –2% to +10% relative to DCM and from –3% to +2% relative to EtAc). These results suggest that Hex/Ace is the most suitable solvent for Soxhlet extraction of the sediment sample. Thus, we examined the time dependence of the extraction efficiency with Hex/Ace as the extraction solvent.

There were no marked increases in either the efficiencies of extraction of native PAHs or the recovery yields of ^{13}C -PAHs (0–17% for native PAHs and 2–11% for ^{13}C -PAHs), although the extraction time was increased from 4 to 24 h. This result suggests that increasing the extraction time will not produce any marked improvement in the efficiencies of extraction of either native PAHs or ^{13}C -PAHs. Because Soxhlet is usually performed within 24 h in most studies [7,8,10,12,14], we believe that the efficiency of extraction of PAHs will plateau after 8 h. Thus, we performed Soxhlet with Hex/Ace for 16 h.

3.2. Investigation of solvents, temperatures, and times for microwave-assisted extraction

MAE is a recently developed technique for extracting organic compounds from solid samples. It requires a shorter extraction time (3–30 min) and consumes less solvent (10–40 mL)

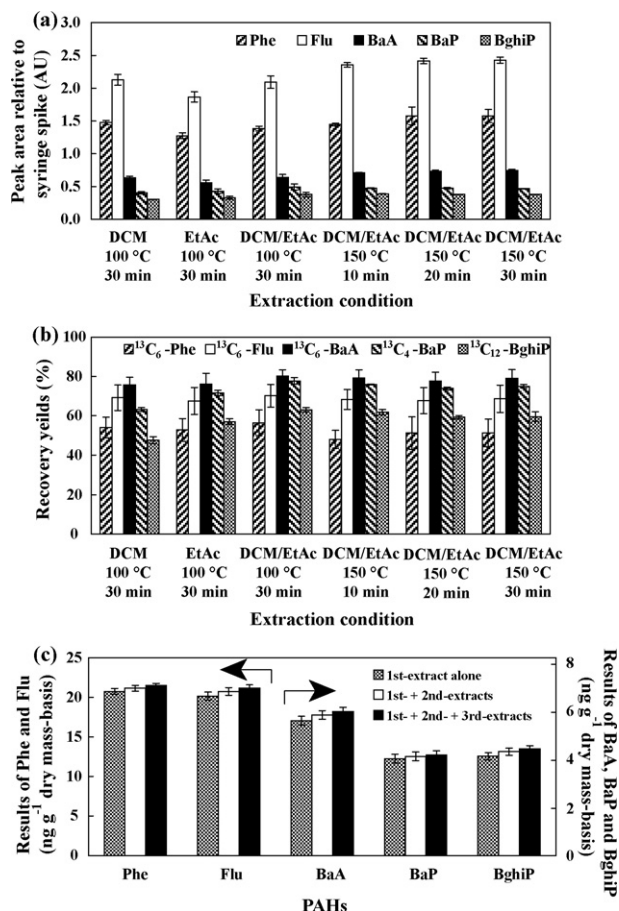


Fig. 2 – Efficiencies of extraction of native PAHs (a) and recovery yields of ^{13}C -PAHs (b) by MAE (DCM, EtAc, and DCM/EtAc at 100 °C for 30 min, or DCM/EtAc at 150 °C for 10, 20, or 30 min), and results (c) by repeated MAE with DCM/EtAc (at 150 °C for 30 min). Error bars indicate the standard deviation ($n = 3$).

than Soxhlet (3–48 h and 100–500 mL, respectively) [24]. Because of these advantages, it has been applied to the extraction of PAHs from solid samples such as sediment, soil, particulate matter, and animal tissue [24]. Efficiencies of extraction of PAHs by MAE are dependent on the solvent, temperature, microwave energy, and time [24]. Since the microwave energy was set at 1200 W and we used ramp-to-temperature mode, we investigated the effects of the solvent, temperature, and time on the extraction efficiencies.

Fig. 2 shows the efficiencies of extraction of native PAHs and the recovery yields of ^{13}C -PAHs achieved with DCM, EtAc, and DCM/EtAc at 100 °C for 30 min, and with DCM/EtAc at 150 °C for 10, 20, and 30 min. Efficiencies of extraction of native PAHs with DCM/EtAc were comparable to or better than those achieved with other solvents (–7% to +25% relative to DCM and +3% to +10% relative to EtAc). On the other hand, recovery yields of ^{13}C -PAHs, especially $^{13}\text{C}_4$ -BaP and $^{13}\text{C}_{12}$ -BghiP, with DCM/EtAc were 1–25% higher (7% higher for $^{13}\text{C}_4$ -BaP and 25% higher for $^{13}\text{C}_{12}$ -BghiP) than those obtained with other solvents. These results suggest that, of the solvents examined, DCM/EtAc is the most suitable for MAE.

With DCM/EtAc as a solvent, we examined the dependences of the efficiencies of extraction of native PAHs and the recovery yields of ^{13}C -PAHs on the extraction time at 150 °C. As shown in Fig. 2a, efficiencies of extraction of native PAHs increased by up to 20% at 150 °C for 30 min compared with the values at 100 °C for 30 min. On the other hand, differences among the efficiencies of extraction of native PAHs performed for 10, 20, and 30 min were less than 9%. These results suggest that 150 °C is better, and no marked improvement relative to increase of temperature in the efficiencies of extraction of both native PAHs and ^{13}C -PAHs should be expected at 150 °C. Because the conditions examined in this study were comparable to or harsher than (in terms of both temperature and time) those applied in previous reports involving MAE [24], we selected DCM/EtAc at 150 °C for 30 min as the extraction conditions for MAE.

The efficiency of extraction could increase as the number of iterations increases. Fig. 2c shows the cumulative values obtained by the first, second, and third extractions. The values did not increase markedly on repetition of the extraction (<8% between the first extract alone and the cumulative values in the second and third extracts). Thus, a single extraction is sufficient. Therefore, we performed MAE once with DCM/EtAc at 150 °C for 30 min.

3.3. Investigation of temperature and extraction cycles for pressurized liquid extraction

PLE has been developed for the extraction of organic compounds from solid samples especially for sediment, soil, and particulate matter samples [9,11,15,16]. Since PLE gives not only comparable extraction efficiency to Soxhlet extraction but also shorter time and less consumption of solvent than Soxhlet extraction, this technique has also been applied to the extraction of PAHs [9,11,16]. Efficiencies of extraction of PAHs using PLE are dependent on the solvent, temperature, and time (or the number of extraction cycles) [9,11]. As for the solvent, we previously examined three solvents such as toluene, DCM, and Hex/Ace mixture (1/1, by volume), and toluene was the most effective solvent for the extraction of PAHs from the same sediment sample [22]. Thus, in this study, we investigated the temperature and the number of extraction cycles on the extraction efficiencies.

Fig. 3 shows the efficiencies of extraction of native PAHs (a) and the recovery yields of ^{13}C -PAHs (b) achieved with toluene at 100 or 150 °C under 15 MPa (2 cycles). Efficiencies of native PAHs at 150 °C were 0.5–38% higher than those at 100 °C, although the recovery yields of ^{13}C -PAHs were comparable (54–92% at 100 °C and 56–84% at 150 °C). This result suggests that PLE at 100 °C is enough for extraction of spiked ^{13}C -PAHs but is not enough for extraction of native PAHs from the sediment sample.

After extraction temperature being set at 150 °C, we investigated the number of extraction cycles on the values. Fig. 3c shows the cumulative values obtained by the first, second, and third cycles. The cumulative values in the first, second and third extracts did not increase markedly from those in the first and second extracts (4.8% for Phe, 0.13% for Flu, 2.1% for BaA, 2.4% for BaP, and 1.1% for BghiP). Thus, we considered that two cycles is enough for PAHs extraction at this tempera-

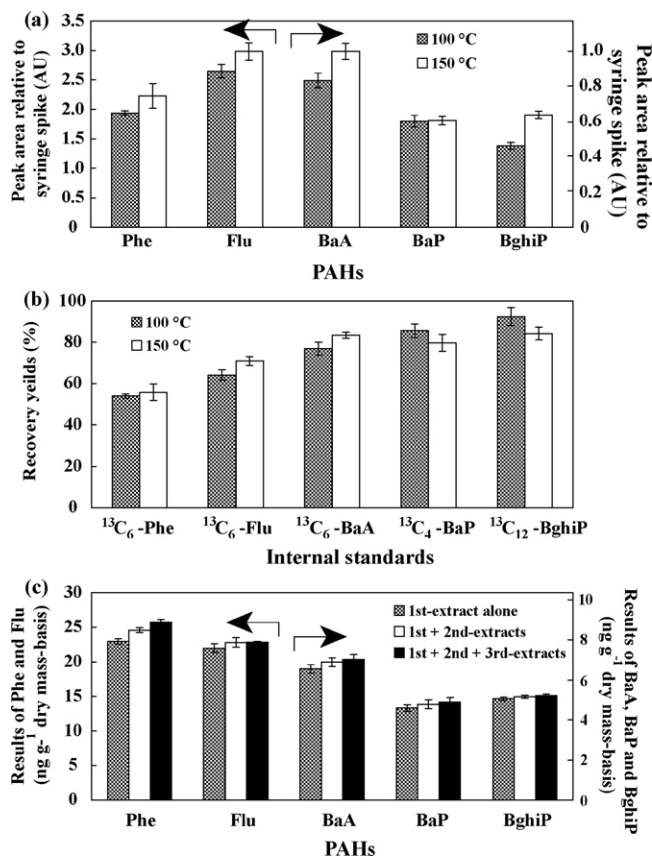


Fig. 3 – Efficiencies of extraction of native PAHs (a) and recovery yields of ^{13}C -PAHs (b) by PLE with toluene at 100 or 150 °C (15 MPa, 10 min), and results (c) by repeated PLE with toluene at 150 °C (15 MPa, 10 min). Error bars indicate the standard deviation ($n = 3$).

ture. Therefore, we performed PLE with toluene at 150 °C under 15 MPa for 10 min (2 cycles).

3.4. Efficiencies of extraction by different techniques

If different extraction techniques under optimized conditions produce the same results, the results can be considered reliable. We compared the results of the extraction techniques. Fig. 4 shows the results of native PAHs and the recovery yields of ^{13}C -PAHs obtained by using Soxhlet, MAE, and PLE. Although all techniques show good repeatability for the five PAHs examined (relative standard deviation <5.2%, $n = 3$), the results from the three techniques show obvious differences and decrease in the order PLE > MAE > Soxhlet; the PLE results were 1.2–1.7 times the Soxhlet results. Because these trends were common among the five PAHs, they are dependent on the extraction techniques or solvents rather than on the characteristics of the particular PAHs (Table 1).

On the other hand, compared with the efficiencies of extraction of native PAHs, the recovery yields of spiked ^{13}C -PAHs were similar among the techniques examined, except in the case of $^{13}\text{C}_{12}$ -BghiP by MAE (perhaps because of re-adsorption on the matrix; see below). Because these recovery yields were also similar to those achieved during clean-up [21],

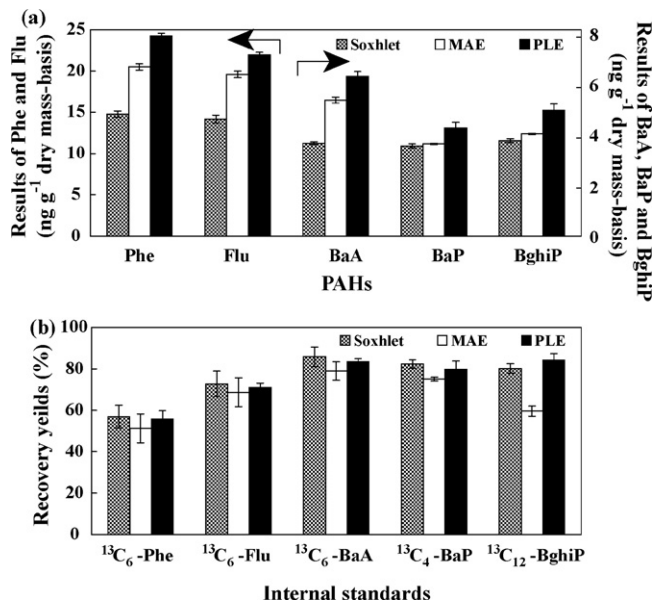


Fig. 4 – Results of PAHs (a) and recovery yields of ^{13}C -PAHs (b) by Soxhlet (with Hex/Ace for 16 h), MAE (with DCM/EtAc at 150 °C for 30 min), and PLE (with toluene at 150 °C under 15 MPa). Error bars indicate the standard deviation ($n = 3$).

all the extraction techniques are capable of extracting most of the spiked ^{13}C -PAHs. The different efficiencies of extraction of native PAHs and ^{13}C -PAHs suggest that spiked ^{13}C -PAHs are more easily extracted than native PAHs. The strength of adsorption of PAHs increases with time after spiking. Eschenbach et al. reported that more than 20% of spiked Flu and BaP could not be extracted with acetone after 56 days due to increase of strength of adsorption to soil [25]. Moreover, Northcott and Jones reported that the strength of adsorption of PAHs to matrix continued to increase beyond 500 days [18]. Thus, an overnight delay between spiking and extraction would be insufficient to achieve an adsorption strength comparable to that of native PAHs.

We determined values of native PAHs by isotope-dilution mass spectrometry (IDMS), which is one of the most reliable methods for precise determination. Because the values were calculated from the ratios of native PAHs to spiked ^{13}C -PAHs and were not dependent on recovery yields of spiked ^{13}C -PAHs, differences in the results between techniques depend mainly on differences in the efficiencies of the techniques at extracting native PAHs. Even if differences in the results between techniques originate from differences in the strengths of adsorption of native PAHs and ^{13}C -PAHs, it is problematic to wait until the spiked ^{13}C -PAHs achieve comparable strengths of adsorption to those of native PAHs. Therefore, the extraction technique should extract not only spiked ^{13}C -PAHs, but also native PAHs, except when isotope equilibrium is complete.

As another explanation for the difference in results among extraction techniques, co-extracted compounds could affect the target peaks of chromatograms. Fig. 5 shows typical total ion chromatograms of extracts obtained by PLE, MAE, and Soxhlet. There are larger number of peaks in the PLE extract than those in the MAE and Soxhlet extracts, but they are earlier

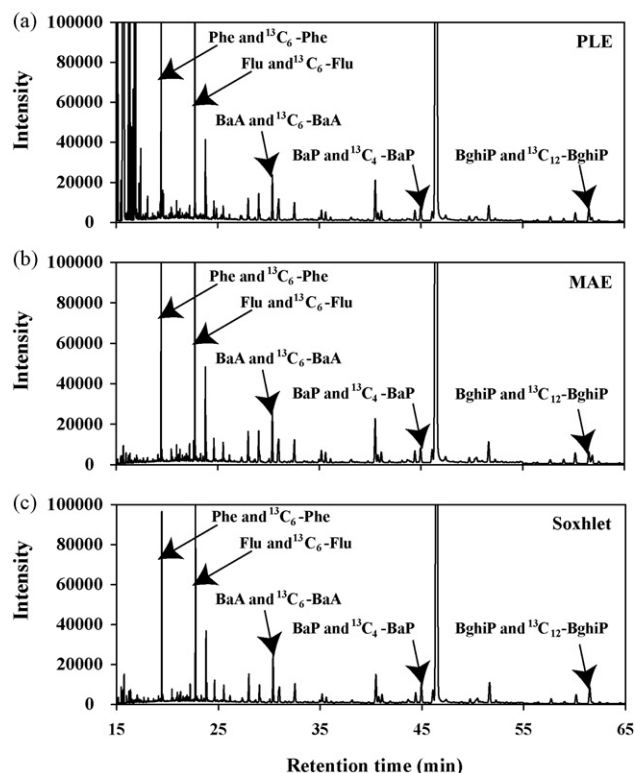


Fig. 5 – Typical total ion chromatograms of extracts obtained by PLE (with toluene at 150 °C, 15 MPa) (a), MAE (with DCM/EtAc at 150 °C for 30 min) (b), and Soxhlet (with Hex/Ace for 16 h) (c).

than peaks of Phe and $^{13}\text{C}_6$ -Phe. But since most peaks used for determination were not affected by co-extracted compounds, the influence of co-extracted compounds on PAH peaks in the chromatograms was negligible.

3.5. Soxhlet extraction followed by pressurized liquid extraction

The results obtained by using Soxhlet and PLE were significantly different (20–72%, $p < 0.05$, Fig. 4a), although Soxhlet is widely used [15,16,23]. To examine the cause of differences in the extraction efficiencies of the two techniques, we performed PLE after Soxhlet on the same sample. Since it was difficult to transfer the whole residue from the Soxhlet thimble to the PLE cell, we performed Soxhlet using a smaller cellulose thimble (20-mm i.d. $\times$ 35-mm long), which we placed inside the usual cellulose thimble (28-mm i.d. $\times$ 100-mm long) for subsequent PLE.

As shown in Fig. 6, the results differed between Soxhlet alone (first extract) and Soxhlet plus subsequent PLE (first+second extracts). When the values are calculated from Soxhlet+PLE (native PAHs and ^{13}C -PAHs in the first+second extracts), they reach those obtained by using PLE alone (a 17–74% increase relative to Soxhlet alone, Fig. 4a). There are two possible explanations for the differences in the results between Soxhlet with Hex/Ace and PLE with toluene (Fig. 4a): the effect of the extraction solvent and the effect of the extraction technique itself, including the operating temperature

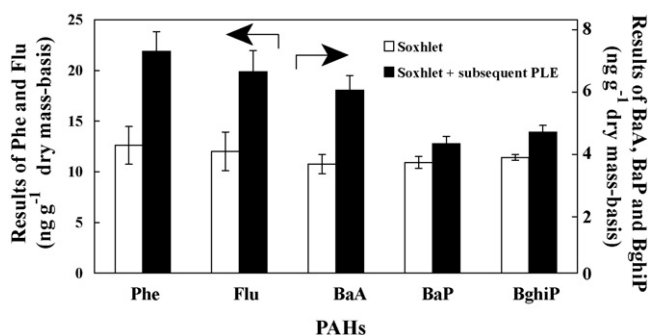


Fig. 6 – Results of PAHs by Soxhlet alone (with Hex/Ace for 16 h) and cumulative results by Soxhlet + subsequent PLE (with toluene at 150 °C, 15 MPa). Error bars indicate the standard deviation ($n = 3$).

and pressure. We previously reported the optimization of the extraction solvent for PLE, and achieved better efficiencies of extraction of the five PAHs with toluene than with Hex/Ace on the same sediment sample; however, the differences were within 5–16% for all five PAHs [22]. Thus, differences in the efficiencies of extraction of the five native PAHs can be attributed to differences in the technique rather than to the extraction solvent.

3.6. Factors introducing different results and recovery yields between PLE and MAE

The results obtained by MAE were closer to those by PLE than to those by Soxhlet; however, significant differences (11–23%, $p < 0.05$, Fig. 4a) were found. Among the ^{13}C -PAHs, the differences in the recovery yield of $^{13}\text{C}_{12}$ -BghiP were remarkable. Two factors could explain these differences: the pressure during extraction (15 MPa in PLE vs. < 2 MPa in MAE), and the different extract-collection steps after extraction between PLE (continuous) and MAE (discontinuous) (see Section 2). To confirm that difference in the solvent used does not affect on the results and recovery yields, we also performed PLE with DCM/EtAc.

Fig. 7 shows the efficiencies of extraction of native PAHs, the recovery yields of ^{13}C -PAHs, and the results obtained by PLE with toluene and DCM/EtAc and by MAE with DCM/EtAc. Most efficiencies of extraction of both native PAHs and ^{13}C -PAHs obtained by PLE with DCM/EtAc were closer to those obtained by PLE with toluene than to those by MAE with DCM/EtAc. This result also supports the view that differences in the results among extraction techniques can be attributed to differences in the extraction techniques themselves rather than to differences in the solvents used. Since the efficiencies of extraction of native PAHs with DCM/EtAc increased in PLE, the results also increased (by 2–14%).

Recovery yields of ^{13}C -PAHs obtained by using MAE were 3–25% lower than those obtained by using PLE, even though the same solvent (DCM/EtAc) was used. As shown in Fig. 4b, recovery yields of ^{13}C -PAHs obtained by MAE were even lower than those by Soxhlet for all the PAHs examined, although there are no reports of such differences in the case of high-level samples [12,13,24]. In our results, similar recovery yields

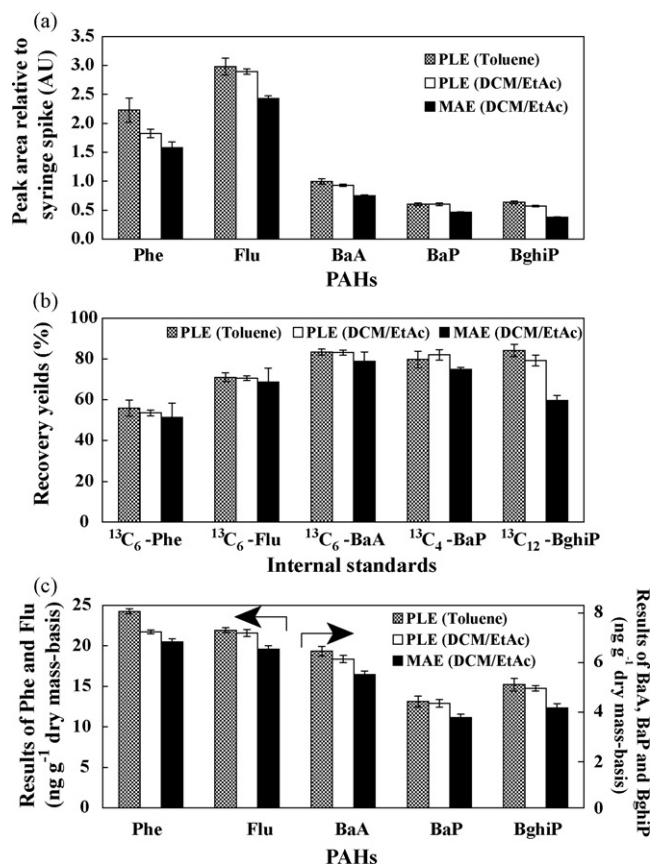


Fig. 7 – Efficiencies of extraction of native PAHs (a), recovery yields of ¹³C-PAHs (b), and results (c) by PLE with toluene (at 150 °C, 15 MPa), PLE with DCM/EtAc (at 150 °C, 15 MPa), and MAE with DCM/EtAc (at 150 °C for 30 min). Error bars indicate the standard deviation (n = 3).

of ¹³C-PAHs by PLE were obtained using DCM/EtAc to those using toluene (Fig. 7b). Differences in the results were less than 8% between the first extract alone and total extracts (first + second + third extracts) when repetition of MAE was examined (Fig. 2c). The efficiencies of extraction of both native BghiP and ¹³C₁₂-BghiP by MAE with DCM/EtAc showed the largest differences from those obtained by PLE with DCM/EtAc (33% for native BghiP and 25% for ¹³C₁₂-BghiP); BghiP has the highest octanol–water partitioning coefficient (log *K*_{ow} value, Table 1), which is a major factor in adsorption to matrix [26]. These results suggest that isotope equilibrium between matrix and solvent can mostly be achieved once, and lower recovery yields of ¹³C₁₂-BghiP by MAE are the result of re-adsorption on the matrix rather than lower extraction efficiency. In terms of determination of PCBs, Numata et al. observed lower recovery yields of ¹³C-PCBs by MAE in a low-level certified reference material (0.15–4.0 μg kg⁻¹-sediment, NMIJ CRM-7305a), but no significant differences in a high-level certified reference material (0.31–61 μg kg⁻¹-sediment, NMIJ CRM-7304a) [15]. Thus, re-adsorption of target compounds in sediment samples at low levels might not be a rare case.

To understand the effect of pressure during extraction, Schantz et al. examined changes in the results of five PAHs in a sediment sample (SRM 1944) by using PLE at 6.9–15.2 MPa:

the values increased with increasing pressure up to 13.8 MPa [11]. On the other hand, the pressure during our extraction using MAE was only 1 MPa (with 30 mL DCM/EtAc at 150 °C). Although it is impossible to confirm that complete isotope equilibrium can be achieved in the PLE cell, there are some reports that higher pressure during extraction more effectively infiltrates solvents into the matrix [27,28]. Since native PAHs are adsorbed to matrix more strongly than ¹³C-PAHs, as mentioned above, the lower results obtained with MAE than with PLE can be attributed to lower pressure during extraction. Since significant amounts of both native PAHs and ¹³C-PAHs were re-adsorbed on the matrix in MAE, separation of the extract from the residue at high temperature and high pressure when using PLE is also important for obtaining higher efficiencies of extraction of both native PAHs and ¹³C-PAHs.

4. Conclusion

In this study, we examined differences in the extraction of PAHs obtained by using three extraction techniques on a sediment sample with low-level contamination. Differences in the results among extraction techniques were mainly attributable to different efficiencies of extraction of native PAHs. Although efficiencies of extraction of PAHs are also dependent on matrix characteristics, it is impossible to examine all matrices owing to their variety (extraction efficiencies must also be dependent on organic content and constituents). However, to evaluate the appropriateness of analytical procedures, not only recovery yields of internal standards but also efficiencies of extraction of native PAHs should be examined.

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